RANDOMIZED SAFETY INSPECTIONS AND RISK EXPOSURE ON THE JOB: QUASI-EXPERIMENTAL ESTIMATES OF THE VALUE OF A STATISTICAL LIFE[1]

by

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CES 14-05 January, 2014

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Abstract

Compensating wages for workplace fatality and accident risks are used to infer the value of a statistical life (VSL), which in turn is used to assess the benefits of human health and safety regulations. The estimation of these wage differentials, however, has been plagued by measurement error and omitted variables. This paper employs the first quasi-experimental design within a labor market setting to overcome such limitations in the ex-tant literature. Specifically, randomly assigned, exogenous federal safety inspections are used to instrument for plant-level risks and combined with confidential U.S. Census data on manufacturing employment to estimate the VSL using a difference-in-differences framework. The VSL is estimated to be between $2 and $4 million ($2011), suggesting prior studies may substantially overstate the value workers place on safety, and therefore, the benefits of health and safety regulations.

Keyword*:* value of a statistical life, hedonic wage models, OSHA, quasi-experiment

JEL Classification: Q58, J17, I18

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1. Introduction

The economic justification for regulations that affect human health and safety relies critically on the monetized value of risk reductions. In an effort to obtain revealed preference estimates of the value individuals place on reducing mortality risks, a large empirical literature has developed

that employs hedonic wage models to estimate the value individuals place on reducing hazards faced at the workplace. Such estimates are used to compute the "value of a statistical life" (VSL),[2] which in turn is used in federal regulatory impact analyses to monetize the benefits of life-saving policies. Despite decades of empirical research, the credibility of VSL estimates obtained from hedonic wage models continues to be the subject of considerable debate (e.g., Black, Galdo and Liu 2003; U.S. OMB 2003; Ashenfelter and Greenstone 2004; Robinson 2007; Cameron 2010; U.S. EPA 2010; Cropper, Hammitt and Robinson 2011). The ongoing academic and policy interest in finding reliable estimates for the VSL is due, in no small part, to the remarkably large role that the VSL has in determining benefit-cost ratios for many federal policies. A recent review of the benefits and costs of 115 major federal regulations promulgated over the past decade indicates that up to 70% of the total benefits across all rules considered are directly attributable to the monetized value of reducing early mortality (U.S. OMB 2013).[3] These benefits are computed by multiplying the estimated number of lives saved as a result of the regulation by an agency's preferred point-estimate for the VSL.

Hedonic wage studies focused on estimating the VSL typically employ cross-sectional or panel data models to estimate compensating wage differentials associated with increased occupational risk.[4] VSL estimates based on these data typically suffer from endogenous

[2] To see how the VSL is computed, suppose there is a group of 100,000 individuals at risk of death from a certain exposure, and it is estimated that the average willingness to pay is $30 per year to reduce the risk of death by 1/100,000. The VSL in this context is equal to $30 x 100,000, or $3,000,000. The VSL does not measure the value of an identified life, but is instead an aggregate of the affected individual's marginal willingness to pay for marginal reductions in risk.

[3] U.S. OMB (2013) reports aggregated benefits and costs for 115 of 536 major rules promulgated between 1992 and 2012 (major rules are defined as those having an economic impact of at least $100 million in any one year). The OMB reports that many of the 536 rules reviewed were budgetary transfer rules and as such, the 115 rules analyzed likely represent a large portion of rules that impose substantial regulatory costs on the private sector, and thus are subject to intense scrutiny.

[4] See Viscusi (1992) and Bockstael and McConnell (2007) for a review of the theory and empirical approaches in the hedonic wage literature estimating compensating wage differentials for occupational risks. See Mrozek and Taylor (2002), Viscusi and Aldy (2003), Kochi, Hubbell and Kramer (2006), and U.S. EPA (2010) for quantitative reviews of the past empirical literature.

regressors and an inability to measure risks at the place of employment. Specifically, risk rates are measured at the national level and are always aggregated by coarsely defined industries and occupations, and are thus subject to considerable measurement error (Black, Galdo and Liu 2003; Black and Kniesner 2003; Scotton 2013). In addition, unobserved worker and job characteristics are likely correlated with job risks and wages, biasing compensating wage estimates in an unknown direction (see Garen 1988; Viscusi and Hersch 2001; Black, Galdo and Liu 2003; Scotton and Taylor 2011 for discussions). Recently Kniesner, et al. (2012) use improved data to estimate panel models that control for unobserved worker characteristics, however the data are not rich enough to address unobserved job characteristics. Because national risk rates are aggregated by broad occupations within industry clusters, identification of the wage/risk premia in these panel models relies on individuals who change jobs and thus change all other job characteristics including risk.

This research is the first to estimate the VSL by employing a quasi-experimental design within a labor-market setting that credibly controls for endogeneity and reduces noise in the measurement of workplace risk.[5] We overcome endogeneity concerns by exploiting conditionally random plant inspections conducted by the federal Occupational Safety and Health Administration (OSHA) to instrument for plant-level risk. OSHA requires firms to correct safety violations within 30 days and follow-up inspections are conducted to ensure compliance. Therefore, these inspections induce exogenous changes in plant-level perceived risks that are likely salient to workers because of reporting requirements.[6] We overcome measurement error

[5]Ashenfelter and Greenstone (2004) estimate the VSL in a quasi-experimental framework outside of the labor-market context by exploiting changes in state speed limit policies.

[6] See Scholz and Gray (1990; 1993), and Gray and Mendeloff (2004) for empirical evidence on the impact of OSHA inspections on plant-level safety.

problems by exploiting 20 years of plant-level fatality, wage, and worksite characteristics data from OSHA and the U.S. Census Bureau.

Our difference-in-differences estimates indicate that production workers' wages are reduced by an average two-to-three percent following inspections, suggesting a range for the VSL of $2 to $4 million in 2011 dollars. These estimates are stable across a wide variety of models and specifications and are 60 to 70 percent lower than the reported range from conventional hedonic wage models of $5 to $13 million, respectively (e.g., Viscusi 2004; Evans and Smith 2008; Hersch and Viscusi 2010; Kniesner, et al. 2012). Our estimates also represent a substantial reduction in the VSL that would be deemed appropriate for federal regulatory impact analyses. For instance, guidance documents for the U.S. Environmental Protection Agency (2000) and U.S. Department of Transportation (2013) indicate that a VSL of approximately $9 million is to be used for their agency regulatory impact analyses moving forward, more than double our highest estimate.

The conclusions drawn from our quasi-experimental estimates are not likely to be purely a result of using a unique new dataset. First, the VSL range reported from conventional models includes estimates arising from similar samples of workers as ours (e.g., Viscusi 2004) and a similar time frame as our labor-market data (e.g., Scotton and Taylor 2011; Kniesner, et al. 2012). Second, we estimate traditional cross-sectional hedonic models using our data merged to national, industry-level fatality risks constructed in a manner consistent with the existing hedonic wage literature. We are able to generate VSL estimates both in the range suggested by the extant literature and in the range of our IV estimates. However, the cross-section models are highly unstable and specification choices play a large role regarding any conclusions drawn from these

models. As a whole, these results underscore that endogeneity and measurement error are likely to have introduced non-trivial biases into conventional VSL estimates.

The remainder of this paper is as follows. In Section 2 we provide a brief historical overview of OSHA practices for selecting plants for safety inspections and identify the periods in which OSHA purports to use a neutral selection process, thus providing the natural experiment we wish to exploit. Section 3 presents an overview of the data and Section 4 provides evidence that OSHA did indeed follow a neutral selection process for inspecting firms during periods in which stated policy indicated random selection procedures be followed. Section 4 also establishes that OSHA inspections affect plant-level fatality rates, the second key condition if inspections are to serve as a relevant instrument for plant-level risks. Section 5 presents estimates of the compensating wage differential associated with plant-level safety improvements and computes point estimates for the VSL. Section 6 offers conclusions.

2. The OSHA Inspection Process

OSHA was established under the Occupational Safety and Health Act of 1970 to set and enforce workplace safety standards. Over the past forty years, the majority of OSHA's funding has been devoted to enforcement of standards through workplace inspections (MacLaury 1984; Siskind 1993; Fleming 2001). Under the Occupational Safety and Health Act, states are able to choose whether they develop and operate their own safety programs or allow the OSHA to operate within the state. This research focuses on the 28 states and the District of Columbia that operate under the Federal OSHA inspection program since a common and transparent scheduling system for conducting inspections is available for these states. Federal OSHA program states are concentrated in the midwest, south and northeastern census regions, and although not a complete

census of U.S. manufacturing plants, the federal states cover 9.5 million manufacturing workers in over 200,000 plants, representing approximately 57% of the U.S. manufacturing workforce.[7]

During its first thirty years, OSHA conducted approximately 24,000 inspections annually in the manufacturing industries. Approximately half these inspections are "programmed inspections," which are comprehensive, unannounced (surprise) inspections. During a programmed inspection, an OSHA compliance officer reviews on-site plant-level records of historical injuries and illnesses, meets with employees, and inspects all aspects of a plant's physical operations that relate to safety or health. Upon completion of the inspection process, the compliance officer holds a closing conference with the employer, employees and/or the employees' representative to discuss any findings. If violations are found, a process begins by which they are categorized by their severity/nature, penalties or fines are established, and OSHA then continues to monitor the plants until all violations are corrected. Most violations are required to be corrected within 30 days, and follow-up inspections are conducted to ensure compliance.

Beginning in 1978, OSHA implemented a scheduling system for selecting plants to receive programmed inspections that targeted high-risk manufacturing industries (defined at the 4-digit SIC level prior to 1999), but randomly selected firms within the industries using a neutral selection criteria at the state-level.[8] Plants with fewer than 11 employees and plants that received comprehensive inspections in the recent past were exempted. The definition of "recent past" for inspection varied between one and three years over our study period. The exemption period

[7] There is not a statistically significant difference in the mean manufacturing employment, wages, and fatality rates among states with federally-administered programs and those with state-administered programs. However, states with federal OSHA inspection programs have mean injury accident rates that are 30% lower than states with state-administered programs. While we recognize that the sample is selected, it is not possible to incorporate state-administered OSHA programs because they are not required to disclose their scheduling processes publicly.

[8] OSHA's inspection scheduling system was developed in response to the 1978 U.S. Supreme Court case Marshall v. Barlow's Inc., which established that OSHA had to obtain a warrant if refused entry to a plant and that probable cause for a warrant could be justified with a documented employee complaint or demonstration that the plant was chosen for inspection through a process of neutral selection (Marshall v. Barlow's Inc. 1978).

policy in force at any given time is defined in OSHA (1981; 1990; 1995). Thus, conditioned on industry, firm size, state and recent inspection history, programmed inspections were to be randomly distributed among plants according to OSHA policy.

Programmed inspections followed this selection criteria from 1978 through 1982, at which point a "records–review" inspection program was established for manufacturing facilities. In this program, plants were still selected for inspection using the same conditionally random selection process as described above. However, upon arrival at a plant, inspectors viewed the plant's illness/injury records and if the plant's rates were below the national average injury/illness rate for all manufacturing industries, the compliance officer would not conduct an inspection. There was a substantial decline in records-review-only inspections beginning in 1984 due to increased criticism that the program created incentives for firms to purposely underreport their injury and illness rates.[9] OSHA responded to the criticism by reducing aggregate numbers of records-review-only inspections as well as adopting a policy of randomly inspecting every 10th plant regardless of their injury/illness rate (Brooks 1988). By late 1987, records-review-only inspections had declined significantly, falling to 6% and 3% in 1988 and 1989, respectively. OSHA discontinued the records review program completely in 1990, returning to the previous scheduling system that randomly selects firms conditioned on industry, state and inspection history. OSHA's conditionally random scheduling system for programmed inspections continued from 1990 until 1998, at which point OSHA substantially changed its selection criteria, moving back to a system that targeted plants based on past injury and illness incidence rates.[10] As a result, this research focuses on OSHA inspections prior to 1999.

[9] See Subcommittee of the Committee on Government Operations House of Representatives (1987).
[10] Beginning in 1999, OSHA targeted plants for inspection whose reported "lost workday and injury and illness rates" in the prior year exceeded a threshold set by OSHA in the current year (OSHA 2004).

Table 1 summarizes OSHA's programmed inspections from 1978 to 1998. Over the twenty year period, there is an average of 24,038 inspections annually in the manufacturing industries. There is an increase in the average annual inspections during the first decade (1978-1987) and a decline over time thereafter. In general, programmed inspections account for the majority of total inspections each year (57%). During the period 1982 to 1987, inspections resulting only in a records review are 26.6% of total programmed inspections, indicating that the majority of plants had injury rates above the national average for all manufacturing and thus received a comprehensive on-site inspection. It is not surprising that records-review-only inspections are a minority of inspections given programmed inspections targeted high-risk industries within manufacturing and the threshold for receiving a comprehensive inspection was the national average injury rate for all manufacturing industries.

In addition to programmed inspections, OSHA also conducts "unprogrammed" inspections that occur in response to an event at the workplace, such as a serious accident, a complaint, a follow-up from a previous inspection to ensure compliance, or for monitoring. While unprogrammed inspections are not the focus of the analysis here, accident inspections are used in the analysis to construct fatality rates at the plant level for the sample of plants. Plant-level accidents that result in a worker fatality, or injuries that require hospitalization of multiple workers trigger an automatic accident inspection.[11] Plants receiving unprogrammed inspections other than follow-up or accident inspections are not included in the analysis because these inspections are not random. This results in approximately 7% of plants being deleted from the sample.

[11] Serious accidents are defined as accidents resulting in a worker fatality or hospitalization of three or more workers (29 CFR Part 1904, 1994). Prior to 1994, it appears that serious accidents were those resulting in a fatality or hospitalization of five or more workers, although we could not identify an exact time frame over which the five-person rule was in place prior to 1994.

Based on OSHA's stated process for selecting plants to receive comprehensive inspections, it is expected that programmed inspections are a plausibly exogenous instrument for plant-level risk during the period 1978 to 1998, with the exception of the period between 1982 and 1987 when the records review policy was in force. In order for OSHA inspections to be a relevant instrument however, they must affect plant-level fatality rates in addition to being exogenous. Empirical tests of the instrument's validity require inspections data to be matched to plant-level data on manufacturing facilities. An overview of this data, as well as the OSHA inspections data, is presented next, followed by empirical tests of the instrument's validity in Section 4.

3. Data

OSHA Inspections

Records from every OSHA inspection conducted between 1978 and 1998 are obtained from the publicly available OSHA Integrated Management Information System (IMIS) inspection database.[12] For each inspected plant, IMIS records the plant name, address, 1987 SIC Classification, date of inspection, type of inspection, found violations, fines levied, plant injury rates for the prior year as recorded in the plant's OSHA-required log reports, and the number of employees at the plant. A census of workplace fatality and serious injuries requiring hospitalization of at least three workers is also available through IMIS because these automatically trigger an OSHA inspection and report.

The OSHA inspection data for each plant is matched to manufacturing facility data available from the Census of Manufactures (COM) using an iterative process that is based on a matching algorithm developed by Fellegi and Sunter (1969). The matching is conducted within two-digit

[12] All data are available at http://ogesdw.dol.gov/raw_data_summmary.php (last accessed April, 2011).

SIC industry classification. A probability matching procedure is implemented which assigns a matching score to each possible record combination in the COM and OSHA inspection data sets. The matching score is the ratio of the probability that a particular sequence of fields are matched given they are true matches to the probability that the sequence of fields are matched given they are *not* true matches. In addition to plant name and address, the comparison space is augmented to include whether or not the two records are a match using the DQMatch function in SAS.[13] A threshold for the matching score is then chosen so that plants above the threshold are considered matches, and plants below the threshold are not. A lower-bound threshold is also established to eliminate records that are almost certainly not matches.[14] As a general rule, about one-half of plants have matching scores above the threshold indicating they are a match. The remaining records are manually checked if their matching score is above the lower-bound threshold. The manual checks are time-intensive, with generally between 100,000 and 250,000 records to inspect per two-digit SIC industry. As such, manual-matching of industries is limited to industries that had relatively high initial match rates as well as having relatively high fatality rates. Although the main analysis focuses on these seven manually matched industries we also present results for all industries.

Table 2 reports the average fatality and injury rates by two-digit SIC code for all manufacturing industries in the five year period 1992 to 1996. Fatalities and injuries are obtained from the Bureau of Labor Statistics' publicly available data.[15] Highlighted in grey are the seven industries that we focus upon in our analysis that include manually matched OSHA

[13] *SAS 9.1.2 Data Quality Server: Reference*, Cary, NC: SAS Institute Inc., 2004

[14] We thank Wayne Gray for sharing code upon which our matching is based. See Scholz and Gray (1990; 1993) and Gray and Mendeloff (2004) for example code implementation using Census and BLS data.

[15] See Table notes for data sources. Fatality data from the Bureau of Labor Statistics' Census of Fatal Occupational Injuries (CFOI). This data is not available prior to 1992 and the coding scheme was changed in 1997 which made it difficult to match fatalities to data on total employment in each industry.

and Census observations. These seven industries account for roughly 50% of all manufacturing establishments. Although some industries such as SIC 24 (Lumber and Wood Products) had high fatality and injury rates, they were excluded from manual matching because of particularly low initial match rates.

Measured in terms of industry fatality rates, the Stone, Clay and Glass (SIC 32), and Primary Metal industries (SIC 33) are the most dangerous industries with average fatality rates of 12.3 per 100,000 workers. Printing and Publishing (SIC 27) is the safest industry with an average fatality rate of 4.3 per 100,000 workers. Table 2 also indicates that there is considerable heterogeneity among industry rankings in terms of fatality rates versus injury rates. Chemicals and Allied products (SIC 28), for example, has the second lowest injury rates of the seven industries, but it ranks third highest in terms of fatalities.

Approximately six to seven percent of the 172,260 plants in the seven industries receive an inspection each year. Table 3 reports the summary statistics for average annual inspections in the seven industries during the study period. The seven industries account for approximately 50% of all manufacturing establishments and 49% of all OSHA inspections conducted in the manufacturing sector. As indicated in Table 3, there are an average of 7,168 inspections per year for the seven industries during the study period. Of the total average annual inspections, 47.6% are the randomly assigned programmed inspections. Fabricated Metals receives the most inspections on average (programmed or unprogrammed), followed by Food Products (SIC 20), Primary Metals (SIC 33) and Stone, Clay, and Glass (SIC 32). It should be noted that the number of inspections in each industry is roughly proportional to the number of plants in each industry, with the Fabricated Metals industry having the most plants among the seven industries.

Table 3 also reports the average number of times a violation is found after a programmed inspection and the average number of violations per inspection. While not directly used in our analysis, violations are the mechanism by which changes in workplace safety conditions are expected to occur. Post-violation, plants are required to correct the issue leading to the violation, usually within 30 days. OSHA also conducts follow-up inspections to ensure the necessary corrections have been made. Across time periods, between 54% and 65% of total inspections result in at least one violation and there are an average of six to ten violations found per inspection. Note, however, when considering only first time inspections, the proportion of inspections resulting in a violation increases to over 75%.

Census of Manufactures (COM)

Beginning in 1967, the COM is conducted every five years by the U.S. Census Bureau and is a complete census of all manufacturing plants in the U.S. Importantly, the COM allows construction of plant-level panel data based on each plant's unique longitudinal identifier, the Permanent Plant Number (PPN). The census contains detailed information on plant-level revenue, expenditures, employment and payroll information. Although data at the plant-level is not released publicly, it is accessed for the purpose of this research through special approval at the Triangle Census Research Data Center.

The COM data reports employment and payroll information for production workers at each plant. Specifically, the number of production workers, total hours worked, and total payroll at each plant are reported, allowing computation of the average hourly wage for production workers at each plant. Being able to distinguish production workers' wages at the plant-level is important as these workers' wages are expected to be more closely linked to plant-level safety since they

are the workers routinely exposed to risks. In addition to payroll information, the COM collects information on the total cost of materials, total revenues, and quarterly number of employees by type of worker. The quarterly employment levels are used to construct a proxy for worker turnover as calculated by the average decrease in total production workers on payroll between quarters across a year as a percent of the average number of production workers employed that year. Table 4 defines the variables available for analysis and provides summary statistics for the seven industries as well as for all manufacturing industries. Data are reported for 1997, which is the most recent COM wave used in the analysis.

As indicated in Table 4, the total value of shipments, number of employees (total and production), total hours worked, and total cost of materials is smaller in the seven industry sample as compared to other manufacturing. These measures indicate plants in the seven industry sample are approximately 70% to 80% the size of plants in the other manufacturing industries. However, worker productivity is nearly equal among the two samples, and the fatality and injury rates are higher in the seven industry sample. Finally, average hourly wage rate for the seven industry sample is $0.40 higher than the average for other manufacturing industries. Appendix Table A1 presents summary statistics for each of the seven industries.

4. OSHA Inspections as an Instrument for Plant-Level Risk

During the contiguous time period 1987 to 1998, OSHA programmed inspections are considered as a plausibly exogenous instrument affecting plant-level risks. To determine if the data support this assumption, it is necessary to test whether OSHA followed stated procedures and the assignment of inspections to plants was indeed random, conditional on the state, industry, and recent inspection history of each plant. Following OSHA guidelines regarding

what type of plants would not be considered for inspection, plants with fewer than eleven employees or which received a comprehensive inspection in the recent past are not included in the analysis.

The following nonparametric matching estimator is used to test for statistically significant differences between the covariates of inspected and uninspected plants:

$$\Delta = E[(X_t^I | industry_i, state_s) - E(X_t^{UI} | industry_i, state_s)], \quad (1)$$

where, covariate *X* in time *t* for an inspected plant (*I*) in industry *i* in state *s* is compared to the average value of covariate *X* in time *t* for all uninspected plants (*UI*) in industry *i* and state *s*. The test statistic, Δ, is the mean difference among differences. The covariates are considered balanced at time *t* if the test statistic is not significantly different than zero at the 10% level.

All observed establishment characteristics listed in Table 4 are tested for balance between inspected and non-inspected groups. Inspections are tracked on an annual basis, while the COM occurs once every five years. As a result, balance is tested in the COM variables among inspected and non-inspected plants in the year the COM takes place, and in each of the four years after the COM wave in question. For instance, using the 1992 COM wave, balance tests are conducted on the 1992 COM variables assigning plants to inspected or uninspected groups based on whether or not they received a programmed inspection in 1992. Balance is also tested in the 1992 COM data assigning plants to inspected or uninspected groups based on whether or not they received a programmed inspection in 1993, 1994, 1995, or 1996. Thus, each wave of the COM data is tested for balance five times using inspection status in each of five years.

The results of these balance tests are presented in Table 5. Covariates for which the null hypothesis is rejected at the 10% level are listed directly in the table. Table 5 highlights the impact of the "records-review-only" policy at OSHA from 1982-1987. In these years, many

covariates fail to balance and this is particularly true between 1982 and 1985 in which OSHA conducted the greatest number of records-review-only inspections. Prior to 1982 and post 1987, however, the vast majority of covariates balance, supporting the randomization assumption during these time periods.

In addition to being exogenous, OSHA inspections must affect plant-level accident rates to be a relevant instrument. This is expected to be the case because post-inspection, all found violations must be corrected and plants are often re-inspected to ensure corrections have been made. OSHA's assignment of violations is the assumed impetus for plant-level safety improvements following inspection, and this supposition is borne out by a number of studies that document plant-level safety improvements after inspection (e.g. Scholz and Gray 1990; Scholz and Gray 1993; Gray and Mendeloff 2004).[16] Nonetheless, a difference-in-differences (DID) estimator is employed to test this assumption that compares fatality rates between inspected and non-inspected plants. Fatality rates are expected to fall post OSHA inspection if inspected plants are indeed coming into compliance with OSHA standards and improving safety at the jobsite.

The inspected plants included in the analysis are limited to only plants receiving randomly assigned programmed inspections and/or unprogrammed accident inspections. While the analysis centers on programmed inspections because of their randomization, unprogrammed accident inspections are also included because these inspections contain the counts of worker fatalities that are necessary for constructing plant-level fatality rates.

The DID estimator is given by:

$$fatrate_{j,t} = a + PC_{j,t}\beta + T_t + P_j + \varphi PI_{j,t} + \epsilon_{j,t}, \quad (2)$$

[16] In a similar vein Viscusi (1986) provides early evidence that OSHA inspections reduce fatality rates at the industry level, although the effect at the aggregated industry level is much smaller in magnitude.

where, the fatality rate (*fatrate*) at plant j in time t is a function of a vector of observable time-varying plant characteristics ($PC_{j,t}$), time and plant fixed-effects, T_t and P_j, respectively, and an indicator variable ($PI_{j,t}$) equal to one if the plant has received a federally programmed OSHA inspection in year *t* or any year prior in the panel.

Equation (2) is estimated using data pooled from three consecutive waves of COM data, supplemented by data from the Annual Survey of Manufactures (ASM) for the years 1987 to 1997. The ASM is a supplement to the COM that collects the same core data in off-COM years for large firms with greater than 1,000 employees and a probability sample of remaining firms based on firm size and contribution to total industry value of shipments.[17] The ASM data are included in the analysis because fatalities are inherently rare events, and exclusion of these data results in a considerable loss of information on annual fatality rates from 1987 to 1997. The constructed panel consists only of plants that receive their first OSHA inspection during the period 1988-1996, and observations for inspected plants are not included for the concurrent period in which the first inspection occurs. There are a maximum of 254,514 observations available for analysis over the study period.

Estimated coefficients from the DID model are presented in Table 6. The first three columns contain estimation results for the seven industries and the last three columns contain results for all industries. Table 6 also presents three alternative estimation strategies for each sample. The first model presented includes plant and year fixed effects as presented in equation (2), the second includes plant, year, industry, and industry-by-year fixed effects, and the final model includes plant, year, industry, and industry-by-state-by-year fixed effects. This latter model follows OSHA's conditionally random selection criteria for scheduling programmed inspections.

[17] Approximately 14% of all manufacturing establishments are surveyed in each ASM. The Census Bureau does not fully disclose the sampling method for including firms in the ASM. Additional details are available at: http://www.census.gov/manufacturing/asm/how_the_data_are_collected/index.html

As indicated in Table 6, there is consistent evidence of reductions in plant-level fatality rates after plants receive an OSHA inspection (see coefficient for "Programmed Inspections", highlighted in bold type). This effect is significant across all samples and stable across models. Note that there is a larger reduction in fatality rates in the seven industry sample as compared to the sample containing all manufacturing industries. Recall that the seven industries are those which have relatively higher fatality rates, so we would expect to see the larger impact of OSHA inspections in these industries. Although the point estimates indicate a larger impact of OSHA inspections in the seven industries, the 95% confidence intervals for the coefficients from comparable models overlap.

Focusing on the results from the samples with industry-by-state-by-year fixed effects, the coefficient estimates indicate that receiving an OSHA inspection reduces the number of fatalities by 2.6 in 10,000 workers in the panel covering all industries. This effect is 35 percent larger for the seven industry sample, with an OSHA inspection reducing the number of fatalities by 3.5 workers in 10,000. On average, in our estimation samples there are 6.1 and 5.0 workers per 10,000 involved in a fatality in the seven industry and all manufacturing industries samples, respectively. Thus, the estimates indicate a reduction in the fatality rate of approximately 52 to 57 percent for plants receiving their first OSHA programmed inspection. While there are no other estimates to which we can directly compare our results, Scholz and Gray (1990; 1993) find that OSHA inspections reduce nonfatal injury rates by 15 to 22 percent. Our larger estimated impact may be due in part to the fact that Scholz and Gray's estimate is an average over all OSHA inspections a plant has ever received, while our estimation strategy focuses on first-time inspections in which the most dangerous violations are likely to be noticed and corrected. Aside from the unreported plant, year, industry, and industry-by-state-by-year fixed effects, there are

generally no other factors that explain changes in a plant's fatality rate other than receiving an OSHA inspection.

Finally, the last row of Table 6 reports F-statistics that compare the unrestricted models reported in Table 6 with restricted models that omit the inspection indicator variable. Staiger and Stock (1997) suggest a cutoff threshold of ten for the F-statistic as a rule of thumb for determining a strong instrument (see also Bound, Jaeger and Baker 1995). Overall, the F-tests suggest the null hypothesis that receiving an OSHA inspection has no significant effect on plant-level fatality rates can be rejected. Furthermore, the F-statistics in the fatality rate equations are all greater than ten, suggesting receiving an OSHA inspection is a valid, strong instrument.

5. Compensating Wage Differentials for Risky Working Conditions

The analysis focuses on randomized inspections as a treatment and assumes that post-inspection, all inspected plants are in compliance with safety rules because they were in compliance to begin with or because they make the changes required by law. The control group to which inspected plants are compared is all uninspected plants. On average, if there are compensating wages for dangerous working conditions as theory would suggest, then one would expect average wages of inspected plants post-inspection to fall relative to the control group. It is not necessary for real wages to fall to identify the wage/risk tradeoff; only that wages rise less quickly at plants whose safety is improving as compared to plants whose safety levels remain unchanged. It is possible that workers are not fully aware of their job risks prior to inspection in which case an OSHA inspection may reveal new safety information to workers and create frictions in the wage adjustment process. The empirical analysis that follows addresses this threat to validity by dropping observations during the year of inspection. Recall that OSHA

violations are required to be corrected within 30 days of violation assignment, and so dropping observations during the year of inspection should allow sufficient time for wage adjustments to occur.

The treatment effect of interest is presented graphically in Figure 1. To construct an event study, the wages of each inspected plant are compared to those of a randomly chosen plant that has never been inspected. Observed wages for inspected plants are drawn from a five-year window before and after the plant received its first OSHA inspection. The matched control plants are restricted to be within the same industry, state, and wage-year as the inspected plant. The new sample of inspected plants and matched controls is then used to estimate the following equation:

$$wage_{j,t} = \sum_{n=-5}^{5} a_n I\left[MIY_{j,t} = n\right] + \sum_{n=-5}^{5} \varphi_n I[PIY_{j,t} = n] + \epsilon_{j,t}, \quad (3)$$

where $MIY_{j,t}$ is a variable equal to the number of years pre/post inspection for inspected plants and their matched uninspected counterpart, $PIY_{j,t}$ is a variable equal to the number of years pre/post inspection for the inspected plants only, $I\left[MIY_{j,t} = n\right]$ is an indicator variable that is equal to 1 if $MIY_{j,t} = n$ and equal to zero otherwise, and $I\left[PIY_{j,t} = n\right]$ is an indicator variable that is equal to 1 if $PIY_{j,t} = n$ and equal to zero otherwise. The coefficient φ_n thus represents the mean difference between inspected and uninspected plant wages n years pre/post OSHA inspection.

The estimated coefficients φ_n and their associated 95% confidence intervals are presented in Figure 1. As depicted in Figure 1 there is a noticeable significant reduction in wages in the first year post inspection, relative to the control group, and each subsequent year through $PIY_{j,t} = 5$. Furthermore, there is not a significant difference in wages between inspected and uninspected

plants prior to the inspection year, which provides further evidence that OSHA inspections are randomly assigned within industry, state, and year.

In order to more precisely estimate the impact of OSHA inspections on plant-level wages, the following difference-in-differences model is estimated:

$$wage_{j,t} = a + \rho\, PI_{j,t} + PC_{j,t}\beta + T_t + P_j + \epsilon_{j,t}, \quad (4)$$

where the average hourly wage rate (*wage*) of production workers at plant j at time t is a function of a vector of five observable time varying plant characteristics ($PC_{j,t}$) including *Turnover*, *Productivity, Number of Employees, Cost of Materials*, *No. Production Workers,* and *Single Unit Plant* (see Table 4 for definitions); year (T_t) and plant-level (P_j) fixed effects. $PI_{j,t}$ is a dummy variable equal to one for plants receiving their first federal programmed inspection in year *t* and each year subsequent to *t*. As such, ρ is the key coefficient of interest that identifies the impact of receiving a federal programmed inspection on wages, or the average treatment effect on the treated. In other words, ρ is the estimate of the compensating differential associated with plant-level safety changes resulting from OSHA inspections. It is expected that ρ will be negative, indicating that improvements in safety at inspected plants lead to reductions in compensation. Finally, $\epsilon_{j,t}$ is a random error term clustered at the plant-level to allow for inter-temporal correlation within clusters.

Equation (4) is estimated using panels of identical length to those used for estimating equation (2) and thus includes three consecutive COM waves from 1987 to 1997. As before, the COM data are supplemented by the ASM, which collects wage and plant-level data annually. Results are robust to inclusion or exclusion of the ASM data. Finally, the wage model is estimated with both the seven industry sample and all manufacturing industries. There are a maximum of 254,514 observations available for analysis. Three specifications are presented

which vary by whether or not industry, industry-by-year, and industry-by-state-by-year fixed effects are included along with plant and year fixed-effects.

Results are presented in Table 7. The estimated coefficient on programmed inspections is negative as expected, and is statistically significant at the 1% level in all samples and specifications. Focusing on the models that allow for industry-by-state specific time trends, the results for the seven industry sample indicate that for plants receiving their first inspection during 1987-1997, the average hourly wages were $0.21 lower post-inspection at inspected plants, or $420 lower per year assuming individuals work an average of 2,000 hours per year.[18] This represents a 2.1% reduction in the average 1997 wages for the seven carefully matched industries. Similarly for the sample covering all manufacturing industries, average wages are $0.20 lower post-inspection, or $400 lower annually (2.0% of average 1997 manufacturing wages).

The estimated sign and significance for the other explanatory variables reported in Table 7 are similar across specifications. Higher production worker turnover rates are associated with higher wages. As one would expect, the coefficient on worker productivity is positive indicating that more productive workers receive higher wages. The variables controlling for plant size have differing impacts on production workers' wages. An increase of 1,000 total employees is associated with a negative albeit statistically insignificant reduction in wage. On the other hand, a $1 million increase in cost of materials is associated with a $0.01 to $0.04 increase in wages. The estimated coefficient for production workers is significant and negative across both samples. An increase of 1,000 production workers is associated with a $2.28 reduction in average wages in the seven industry sample, and a $1.67 reduction in average wages in the full sample. Interestingly, the negative estimated coefficients on total employees and average number of production

[18] In order to allow sufficient time for wage adjustments to occur, observations in which the census year is equal to the year of inspection are dropped. As a result, observations at establishments receiving an OSHA inspection in 1987 or 1997 are dropped from the sample.

workers seem to contradict traditional empirical findings that suggest large firms pay a wage premium in comparison to small firms (see Oi and Idson 1999 for a detailed review). This peculiarity may be due to the fact that the estimation strategy employed in equation (4) makes use of panel data and controls for plant-level fixed effects. Identification is based on within-unit changes in the number of employees. The results are therefore consistent with classical theory in which workers are paid their marginal revenue product. Intuitively, if employment is reduced at the plant, we expect higher marginal productivity for remaining workers which would be associated with higher wages.

Robustness

The estimated impact of an inspection on wages is only consistent if wages at inspected plants followed a common trend with the wages at non-inspected plants prior to inspection. More formally, let D_T be an indicator variable that is equal to one if a plant is inspected in time *T*, and equal to zero otherwise. Assuming a simple two period model where some plants are inspected at time *T* and no plants are inspected prior to *T*, the common trend assumption can be written as:

$$E(wage_T^{IP}|D_T = 0) - E(wage_{T-1}^{IP}|D_{T-1} = 0) =$$
$$E(wage_T^{NIP}|D_T = 0) - E(wage_{T-1}^{NIP}|D_{T-1} = 0), \quad (5)$$

where $wage^{IP}$ is the wage at an inspected plant, and $wage^{NIP}$ is the wage at a non-inspected plant. In equation (5), $E(wage_T^{IP}|D_T = 0)$ is the unobserved counterfactual expected wage rate at the inspected plant if the inspected plant were not inspected in *T*, and $E(wage_{T-1}^{IP}|D_{T-1} = 0)$ is the expected wage rate of the inspected plant in *T-1*, which is prior to receiving an OSHA inspection. Likewise, $E(wage_T^{NIP}|D_T = 0)$ is the expected wage rate of the non-inspected plant during the

post-inspection period, and $E(wage_{T-1}^{NIP}|D_{T-1}=0)$ is the expected wage rate of the non-inspected plant during the pre-inspection time period.

To explore the validity of the common trend assumption the following is estimated:

$$wage_{j,t} = a + PC_{j,t}\beta + T_t + P_j + \rho' PI_{j,t+10} + \epsilon_{j,t}, \quad (6)$$

where all variables are as defined in equation (4), but now wages of plant j in time t are regressed on inspection status of the plant in time t+10. A plant's inspection status at time t+10 is used as a "pseudo" treatment variable in a comparison of wages. Thus, for example, the 1977-1987 panel of wages is regressed on plant characteristics in 1977-1987 and on inspection status as defined by the 1987-1997 panel. If there is a common trend in wages among inspected and uninspected plants, the estimated coefficient ρ' is expected to be statistically indistinguishable from zero.

Results from the estimation of equation (6) are presented in Appendix Table A2 for the seven industry and full sample, respectively. The "pseudo" treatment variable for programmed OSHA inspections is not statistically significant at any conventional level of significance in either equation, failing to reject the common trend assumption for inspections occurring between 1987and 1997. Also, the other covariates in Table A2 generally have the same sign as the estimated coefficients from equation (4), but the magnitudes and statistical significance are somewhat different. Worker productivity, for example, has a much larger highly significant impact on wages during the 1977-1987 period as opposed to the 1987-1997 period.

As a further test of model specification, equation (4) is estimated using the transformed natural log of wages as the dependent variable. Results from the estimation of the log-linear wage equations are presented in Table A3 in the appendix. The results are similar in sign but smaller in magnitude when compared to the linear specification presented in Table 7. An OSHA inspection is estimated to reduce wages by 1.6% in the seven industry sample and 1.5% in the sample

covering all manufacturing industries. This is somewhat smaller than the 2.1% (seven industry sample) and 2.0% (full sample) estimated wage reductions in the linear specification given in equation (4), but the 95% confidence intervals all overlap.

An additional concern was that OSHA inspections may affect wages through other channels such as worker layoffs due to costly fines. We test this hypothesis in two ways. First, we estimate the following:

$$No.\ production\ workers_{j,t} = a + PC_{j,t}\beta + T_t + P_j + \gamma\, PI_{j,t} + \epsilon_{j,t}, \quad (7)$$

where all the variables are defined as in equation (4), except the dependent variable is the average number of production workers at plant j and time t. Although not reported for succinctness, results indicate that contrary to the hypothesis of worker layoffs there is a statistically significant increase in the average number of production workers employed following an OSHA inspection. In addition to estimating the response of production worker employment to OSHA inspections, we conduct a falsification test and estimate the following:

$$other\ employees\ wages_{j,t} = a + PC_{j,t}\beta + T_t + P_j + \gamma\, PI_{j,t} + \epsilon_{j,t}, \quad (8)$$

in which the dependent variable is now the wages of other non-production employees (e.g., clerical and management positions). We expect non-production employees to be unaffected by safety inspections as they are not likely to be exposed to risks associated with manufacturing production technologies. Consistent with our expectations, we find no statistically significant impact at the 5 percent level of OSHA inspections on the wages of non-production workers.[19]

Another implicit assumption of the wage equation (4) is that the treatment effect of receiving a programmed OSHA inspection is persistent across time. Transitory treatment effects may exist if OSHA inspections lead to short-term safety improvements and inspected plants revert back to

[19] In the seven-industry sample, we do observe a statistically significant increase of other employee wages in response to an OSHA inspection at the 10% level. However, this is not the case for the full-sample.

work practices that violate OSHA standards post-inspection. If transitory treatment effects are present, equations (2) and (4) will provide a downward biased estimate of the impact of a programmed OSHA inspection. Transitory treatment effects are not expected in this application because as highlighted in Section 2 plants often receive follow-up inspections to ensure that compliance measures are enacted at violating plants. Furthermore, there are monetary penalties associated with repeated OSHA violations that provide incentives for plants to persistently follow compliance measures addressing past OSHA violations.[20]

We test for transitory treatment effects in two ways. First, we allow treatment effects to vary by each year post-inspection as follows:

$$fatrate_{j,t} = a + PC_{j,t}\beta + T_t + P_j + \sum_{n=1}^{9} \varphi_n I[PIY_{j,t} = n] + \epsilon_{j,t}, \quad (9)$$

$$wage_{j,t} = a + PC_{j,t}\beta + T_t + P_j + \sum_{n=1}^{9} \rho_n I[PIY_{j,t} = n] + \epsilon_{j,t}, \quad (10)$$

where all variables are as defined in equations (2) and (4) except $PIY_{j,t}$ is a count variable equal to the number of years post the initial inspection year and $I[PIY_{j,t} = n]$ is an indicator function that is equal to one for observations that are *n* years post inspection and equal to zero otherwise. If treatment effects are transitory, φ_n and ρ_n will be statistically indistinguishable from zero for larger values of *n*. Although not reported for succinctness, results are that estimates of the coefficients φ_n and ρ_n are always negative, stable in magnitude, and are generally statistically significantly different from zero for all *n*. The results therefore suggest that treatment effects are persistent and the wage and fatality transitions specified in equations (2) and (4) correctly identify the wage and job risk dynamics following a programmed OSHA inspection.

[20] Repeated violations are offenses that are found upon re-inspection that are similar to original infractions, and carry a maximum fine of $70,000 (OSHA 2002).

A second way to explore whether treatment effects are transitory is to examine the changes in the number of times a plant is found to be in violation of OSHA rules during an inspection. To accomplish this, the following model is estimated:

$$violations_{j,t} = a + PC_{j,t}\beta + T_t + P_j + \sum_{n=1}^{10} \varphi_n I[IN_{j,t} = n] + \epsilon_{j,t}, \quad (11)$$

where all variables are as defined in equation (4) except the dependent variable of interest is a count of the total number of violations found at plant j at time t. $IN_{j,t} = n$ is a count variable equal to the inspection number for plants inspected in year t (i.e. if $IN_{j,t}$ is equal to ten for plant j at time t, then plant j received their tenth OSHA inspection at time t). $I[IN_{j,t} = n]$ is an indicator function that is equal to 1 for observations receiving their n^{th} inspection at time t and equal to zero otherwise. Results indicate that the number of violations assigned appears to steadily decrease with additional follow-up inspections. Based on results from the full sample, initial inspections result is 4.5 violations on average (95% confidence intervals lie between 3.4 and 5.6 violations), and plants receiving their tenth inspection receive 1.6 violations on average (95% confidence intervals lie between -0.03 and 3.1 violations).

Measuring the value of reducing workplace risks

To compute the VSL, the average change in wages associated with safety improvements as estimated by ρ in equation (4) is combined with the average change in plant-level fatality rates as estimated by φ in equation (2) as follows:

$$\text{VSL} = {}^{\rho}/_{\varphi} * 2{,}000 * 10{,}000, \quad (12)$$

The average worker is assumed to work 2,000 hours a year, and fatality rates are measured by deaths per 10,000 workers. Since the VSL is a ratio of two OLS estimates, instrumental

variables estimates were used to calculate the robust standard errors necessary for constructing 95% confidence intervals for the Wald estimators of the VSL.

Panel A in Table 8 presents VSL point estimates and 95% confidence intervals based on models using the seven industry sample and the full sample. All estimates are reported in 2011 dollars. The VSL point estimates are between $2.3 million and $4.1 million (95% confidence intervals lie between $0.4 and 6.2 million). Our preferred estimates are $2.31 for the limited sample of seven industries and $3.09 for the full sample because these estimates arise from models that include controls for OSHA's stated criteria regarding conditionally random inspection assignment. These estimates are approximately 60 to 70 percent lower than the reported range of $5 to $13 million from recent hedonic wage studies that are based on the best available national risk rates.[21] Importantly, this range includes estimates arising from samples of only blue collar workers taken during the same time period as our study (e.g., Viscusi 2004 reports estimates of $10 to $12 million for his comparable sample of workers), risk rates that vary only by industry (e.g., Evans and Smith 2008; Evans and Schaur 2010), and risk rates that vary by occupation within industries (e.g., Kniesner, Viscusi and Ziliak 2010; Scotton and Taylor 2011; Kniesner, et al. 2012; Scotton 2013).

Note that underlying the VSL as constructed in equation (12) is a treatment group of inspected plants that contains both violators (whose safety are improving) and compliers (whose safety are unchanged). As a result, the estimated coefficient ρ in equation (4) is an unbiased estimate of the causal effect of an OSHA inspection, but it is a downwardly biased estimate of the compensating wage differential for working at a riskier plant (as evidenced by being in

[21] Reported VSL estimates tend to lie closer to the upper-end of the $5 to $13 million range. A notable exception is Kochi (2011) who estimates panel data models in a similar vein to Kniesner, et al. (2012) using worker data from the Survey of Income and Program Participation (SIPP). Kochi reports VSL estimates of $2 to $3 million. We are not aware of other hedonic wage studies estimating the VSL using SIPP data to which Kochi's results can be compared.

violation of OSHA standards). The VSL, however, is constructed as the ratio of the change in wages following an OSHA inspection to the change in fatality rates following an OSHA inspection. The fatality rate differential following an OSHA inspection is also a downwardly biased estimate of the violator versus complier fatality rate differential, but it is downwardly biased by the same proportion as the compensating wage differential. As a result, the constructed VSL is an unbiased estimate of the local average treatment effect of plant safety on wages for those plants who are incentivized to improve safety following an OSHA inspection.

There is however potential bias in our estimate of the VSL that arises from the fact that injury rates and fatality rates are not simultaneously controlled in our structural wage regressions. If injury rates are positively correlated with wages and fatality rates, and injury rates are negatively correlated with OSHA inspections, the VSL estimated in equation (12) is biased as follows:

$$\text{VSL} = \left(\rho+\frac{\sigma_{PI,IR}}{\sigma_{PI,PI}}\delta\right) \Big/ \left(\varphi+\frac{\sigma_{PI,IR}}{\sigma_{PI,PI}}\gamma\right) *2{,}000*10{,}000, \tag{13}$$

where , $\sigma_{PI,IR}$ is the covariance between programmed inspections and injury rates, $\sigma_{PI,PI}$ is the variance of programmed inspections, and δ and γ are the true parameters describing the impact of injury rates on wages and fatalities, respectively. All other variables are defined as in equation (12). The direction of bias is determined by the relationship between δ and γ. While we cannot identify this relationship with certainty, back of the envelope calculations suggest that our estimates of the VSL represent an upper-bound. Current research exploring the relationship between wages and non-fatal injuries suggest that a 1/10,000 increase in worker injury risk is associated with a \$0.002 - \$0.004 increase in the wage rate, or δ (Viscusi and Aldy 2003). Based on a simple regression of aggregate injury rates on fatality rates using the same data as described

in Table 2, the estimated coefficient γ is predicted to be an order of magnitude smaller than the coefficient δ, suggesting our VSL estimates are likely upper-bound estimates. Note, many studies using modern national workplace fatality statistics also exclude injury rates from their empirical analysis (e.g., Viscusi 2004; Evans and Schaur 2010; Kniesner, Viscusi and Ziliak 2010; Kniesner, et al. 2012; Scotton 2013). When the matter is discussed, the authors indicate that inclusion of injury risks either lowered their VSL point-estimates (Viscusi 2004) or did not affect their estimates appreciably (Kniesner, et al. 2012; Scotton 2013). Meta analyses of the older empirical literature (pre-2000) also suggest similar effects of omitting injury rates (Mrozek and Taylor 2002; Viscusi and Aldy 2003).

It is also instructive to compare our IV estimates of the VSL to those obtained from a traditional hedonic wage model estimated with our plant-level data matched to national risk rates typically employed in the literature (Mrozek and Taylor 2002; Viscusi and Aldy 2003 provide an overview of earlier studies; see also Evans and Schaur 2010). Specifically, fatalities at individual plants are aggregated to a national level by 2-digit and 4-digit SIC industry codes. The resulting fatality risks are similar in vein to Viscusi (2004), Kniesner, et al. (2012), Evans and Schaur (2010) and Scotton (2013) who also construct national average risk rates at the 2-digit SIC industry code. While these past studies pool across types of workers and thus also vary risk rates by 10 broad occupational classes within each industry, one of these classes is "Production" and thus our risk rates are directly analogous within the context of our specific sample of workers.

The hedonic wage model estimated is:

$$wage_{j,i,t} = a + PC_{j,i,t}\beta + T_t + I_i + \partial FR_{i,t} + \epsilon_{j,t}, \quad (14)$$

where the average hourly wage rate of production workers at plant j in industry i at time t is a function of a vector of five time-varying plant characteristics ($PC_{j,i,t}$), year fixed effects (T_t), industry-level fixed effects (I_i), and our constructed national average industry-level fatality rates, $FR_{i,t}$. The estimated coefficient ∂ measures the compensating wages associated with a 1/10,000 increase in the probability a job-related fatality. Models are also estimated in which the dependent variable is the natural log of wages. Sample estimates of ∂ and the associated VSL are presented in Panels B and C of Table 8 for wage and log-wage specifications, respectively. The models presented also vary by whether industry fixed effects at the 2-digit SIC level are included. Models that replace industry fixed effects with plant fixed effects are also reported.

As indicated in Table 8, the models including risk rates constructed by 2-digit SIC result in VSL estimates that are either implausibly high (over $75 million) or have a counterintuitive negative sign.[22] Models using fatality rates constructed at the 4-digit SIC level are somewhat more stable. Exclusion of industry or plant fixed effects results in a VSL estimate of approximately $7.0 million, consistent with the range reported in the current hedonic wage literature. However, inclusion of 2-digit SIC industry fixed effects lowers these point estimates to approximately $3.0 million, and inclusion of plant fixed effects again results in a negative, albeit statistically insignificant, coefficient estimate.[23] Thus, it is possible to generate VSL estimates both in the range suggested by the extant literature and in the range of our IV estimates using conventional hedonic wage models applied to our data. These results suggest that the conclusions drawn from our quasi-experimental estimates are not purely a result of using a

[22] Kniesner, et al. (2012) also estimate implausibly large VSL point estimates between $17 million and $29 million using cross-sectional data from the Panel Study of Income Dynamics.
[23] Although not reported in Table 8, models including 4-digit SIC industry fixed effects result in insignificant VSL estimates.

unique new dataset. However, the estimates from our conventional models are suspect given their highly unstable nature and the large role specification choices play in the conclusions.

6. Conclusions

This research provides the first quasi-experimental estimates of the VSL within a labor-market context as an alternative to traditional hedonic wage applications. Notably, the research uses exogenous changes in plant-level risks, which are a much improved alternative to the nationally aggregated measures of workplace risks that are typically used in hedonic wage applications. Furthermore, we are able to ameliorate concerns regarding omitted variable bias by using randomly assigned OSHA inspections as an exogenous instrument affecting plant level safety.

Results from the analysis of all manufacturing industries using the most flexible econometric specification indicates that on average, workers' wages are reduced by 2.0 percent post-inspection, or approximately $792 annually in 2011 dollars. This estimate translates to a point estimate for the VSL of $3.1 million.[24] Across samples and specifications, our VSL estimates range between $2 and $4 million. These estimates are generally 60% or more lower than current VSL estimates computed from conventional cross-sectional hedonic wage models and from current panel models of individual workers that rely on job-switchers to identify a wage/risk

[24] Recently, Kuminoff and Pope (2013) highlight that quasi-experimental applications in hedonic property value studies estimate capitalization rates (differences between hedonic equilibria) under certain conditions rather than identifying marginal willingness to pay (MWTP) measures that are obtained from a single, stable hedonic rent gradient à la Rosen (1974). However, we do not expect this issue to affect the comparison between our quasi-experimental application and past hedonic wage studies attempting to recover MWTP for the following reasons. First, capitalization rates (differences in prices arising from movement among hedonic equilibria) are equal to MWTP under the assumption that changes in the characteristic of interest does not affect the shape or the slope of the hedonic price function (Kuminoff and Pope 2013). We find it unlikely that plant-level OSHA inspections change the shape of the labor market hedonic equilibrium given only six to seven percent of plants are inspected by OSHA each year in our sample. Furthermore, Kuminoff and Pope (2013) show that a capitalization rate identifies MWTP even when the gradient of the hedonic price function changes as long as the instrument is randomized. Our results clearly suggest that OSHA inspections are randomly assigned.

premium. There are few other quasi-experimental studies to which we can compare our results. Most notably is Ashenfelter and Greenstone (2004) who exploit changes in interstate speed limits to compute an upper-bound point estimate for the VSL of $2.2 million (2011$), which is within the range of our estimates.[25]

In addition to contributing to the growing literature exploiting quasi-experimental designs to improve revealed preference evidence for the valuation of nonmarket goods (e.g., Davis 2004; Greenstone and Gallagher 2008; Pope 2008), this work has important implications for the debate surrounding the appropriate VSL for use in federal regulatory impact analyses. From a policy perspective, the VSL is perhaps the most important single nonmarket valuation estimate provided by economists. For example, the guidance issued by two key agencies, the U.S. EPA and U.S. DOT, indicate that agency-wide benefit-cost analyses should use approximately $9 million as the central estimate for the VSL. The evidence underlying these guidance statements are either wholly from the conventional hedonic wage literature reviewed herein in the case of U.S. DOT (2013), or primarily from that literature in the case of the U.S. EPA (2000). Our study implies that the guidance central estimate for the VSL by these two agencies is overstated by threefold.

Taken as a whole, our results suggest that compensating wage differentials for risky working conditions do indeed exist as suggested by theory and explored empirically in the hedonic wage literature for over 40 years. However, our results also suggest that the empirical challenges inherent in estimating the VSL via cross-sectional or panel-data hedonic wage models have not yet been fully addressed. Data limitations, especially with regards to the measurement of risk

[25] In related applications, León and Miguel (2013) exploit exogenous variation in transportation choice fatality risks in Sierra Leone and estimate a VSL of $0.6 million to $0.9 million, while Greenstone, Ryan and Yankovich (2012) estimate a structural model of re-enlistment choices that utilizes exogenous variation in re-enlistment bonuses to estimate a VSL of $3.0 to $4.0 million for military personnel.

faced by workers at their worksite, and correlated unobservables are likely the key impediments to identification of unbiased compensating wage differentials. Of course, our approach is not without data limitations as well. Wages are aggregated at the plant-level and we do not have a representative set of industries. Continued research that carefully identifies exogenous variations in risk that are linked to observable monetary tradeoffs is clearly needed.

Table 1. OSHA Inspections Summary for the Manufacturing Industries.[a]

Time Period	Inspection Process	Total Inspections	Programmed Inspections [% that are records review]	Accident Inspections
		Annual Averages		
1978 – 1981	Conditionally random[b]	28,025	10,353 [2.7%]	861
1982 – 1987	Records-review[c]	37,694	27,726 [26.6%]	800
1988 – 1989	Records-review is phased out	20,158	9,606 [4.7%]	483
1990 – 1998	Conditionally random	14,025	6,686 [0.8%]	408
1999-present	Injury threshold triggers inspection[d]	n/a	n/a	n/a
Total Period: 1978 – 1998		24,038	13,674 [8.9%]	613

[a] All data compiled from the OSHA IMIS data, available as downloadable files from: www.ogesdw.dol.gov/raw_data_summmary.php (last accessed April, 2011).

[b] High-hazard industries are targeted. Plants within these industries are randomly selected within each state conditional on the plant having 11 or more employees and having not received a comprehensive review within the recent past. See footnote 6 (Section 2) for the OSHA's definition of "recent past" when exempting plants.

[c] "Records-review" inspections are conducted. Plants are selected for inspection using the same process as for conditional random inspections. However, upon arrival at a plant the inspector reviews injury/illness logs and only conducts a comprehensive inspection if reported injury/illness rates are above the national average (across all industries).

[d] Plants are selected for inspection based on their reported previous year's injury and illness rate. Reported rates above a predetermined threshold trigger inspection.

Table 2: Average Industry Fatality Rates and Injury Rates, 1992 to 1996.[1]

Industry Name	SIC	Injury Rate[2]	Fatality Rate[3]
Food and kindred products	20	19,788	7.5
Tobacco products	21	9,179	0
Textile mill products	22	9,399	2.3
Apparel and other textile products	23	7,235	1.0
Lumber and wood products	24	17,859	34.9
Furniture and fixtures	25	15,330	2.9
Paper and allied products	26	12,168	5.1
Printing and publishing	27	10,389	4.3
Chemicals and allied products	28	10,679	7.4
Petroleum and coal products	29	9,633	19.1
Rubber and misc. plastics products	30	15,044	3.7
Leather and leather products	31	12,718	0.8
Stone, clay, and glass products	32	17,250	12.3
Primary metal industries	33	20,634	12.3
Fabricated metal products	34	19,068	4.5
Industrial machinery and equipment	35	16,969	4.2
Electronic and other electric equipment	36	10,275	2.0
Transportation equipment	37	23,851	4.9
Instruments and related products	38	8,344	1.9
Misc. manufacturing industries	39	11,376	4.1

[1]The highlighted fields are for the seven manually matched industries that have high fatality rates and had high initial match rates.

[2]Injury rates are computed by dividing the total number of workplace injuries by the total number of employees within the industry, and are scaled to represent injuries per 100,000 employees. Workplace injury count data are obtained from the publicly accessible Bureau of Labor Statistics (BLS) Annual Survey of Injuries and Illness (ASOII), and are scaled to represent injuries per 100,000 employees. The ASOII is available at http://www.bls.gov/iif/data.htm (last accessed May 2012).

[3]Fatality rates are calculated by dividing the total number of workplace fatalities as recorded in the BLS Census of Fatal Occupational Injuries (CFOI) by the total number of employees within the industry. Fatality rates are scaled to represent fatalities per 100,000 employees. The CFOI is available at http://www.bls.gov/iff/data.htm (last accessed May 2012).

Table 3: Average Annual OSHA Inspections by Two-Digit SIC Code, 1988 to 1998.[1]

	Annual Averages			
	Total Inspections (% of seven industry total)	**Programmed Inspections**	**Inspections Resulting in a Violation**	**Violations per Inspection**
		(% of total for column) [% of total inspections for the industry]		
Food Products	1,317 (18.4)	707 (20.7) [53.7]	431 (21.7) [60.9]	7.2
Paper Products	519 (7.2)	207 (6.1) [39.9]	133 (6.7) [64.2]	8.6
Printing	472 (6.6)	155 (1.0) [32.8]	94 (4.7) [60.6]	6.5
Chemical Products	665 (9.3)	191 (5.6) [28.7]	105 (5.3) [54.7]	7.0
Stone, Clay and Glass	773 (10.8)	416 (12.2) [53.8]	241 (12.2) [58.0]	7.7
Primary Metals	947 (13.2)	379 (16.1) [40.0]	235 (11.9) [61.9]	9.8
Fabricated Metals	2,475 (34.5)	1,354 (39.7) [54.7]	744 (37.5) [54.9]	8.5
Total	7,168 (100)	3,409 (100) [47.6]	1,983 (100) [58.2]	7.9

[1]All data compiled from the OSHA IMIS data, available as downloadable files from: www.ogesdw.dol.gov/raw_data_summmary.php (last accessed April, 2011).

Table 4. Average Employment, Payroll, and Plant Characteristics (1997).[1]

Variable Name	Description	Seven Industries	Other Manuf.	All Manuf.
Number of Employees	Annual average of total employees reported per quarter, per plant.	38.5	53.4	46.3
No. Production Workers [% of total employees]	Annual average of employees reported per quarter as production workers, per plant.	28.3 [73%]	37.8 [71%]	33.3 [72%]
Hourly Wage ($1997)	Average hourly wages of production workers in a plant; computed as the total annual payroll for production workers divided by total hours worked.	14.22	13.82	13.99
Production Workers' Payroll	Total annual payroll for production workers, measured in $millions.	0.8	1.0	0.9
Total Hours Worked By Production Workers	Total annual hours worked by production workers, measured in thousands.	57.5	74.9	66.7
Fatality Rate	Average worker fatality rate calculated as the number of fatalities in an industry divided by the number of workers in the industry per 100,000 workers.	6.7	5.7	6.1
Injury Rate	Average worker injury rate calculated as the number of injuries in an industry divided by the number of workers in the industry per 100 workers	14.9	12.6	13.5
Cost of Materials	Total cost of all materials consumed or put into production for the year, measured in $millions.	4.7	6.3	5.6
Total Value of Shipments	Total value of all products shipped by a plant, measured in $millions.	9.2	11.8	10.6
Productivity ($1997)	Total value of shipments divided by production workers' total hours worked.	159.84	157.69	158.57
Single Unit Plant	Dummy variable equal to 1 for plants that are single unit establishment and equal to zero for multi-unit establishments.	81%	82%	82%
Turnover[2]	Average decrease in production workers on payroll between quarters across a year as a percent of the average number of production workers employed that year.	N/A	N/A	N/A

[1] All data are from publicly available COM files for the year 1997 and thus do not match our restricted-access sample upon which the empirical analysis is based. The data is downloadable via http://factfinder.census.gov/home/saff/main.html?_lang=en (last accessed Nov. 2011). Table 2 presents definitions and sources of fatality and injury rate data.
[2] Data on worker turnover rates is not publicly available.

Table 5. Randomization Tests for Federal Programmed Inspections.

COM Wave	**Year of Inspection**	**Variables Failing to Balance**[a]
1977	1977	*0*[b]
	1978	*0*
	1979	*0*
	1980	*Total # Employees, Average # Production Workers*
	1981	*0*
1982	1982	*Total # Employees, Cost of Materials, Single Unit, Productivity, Average # Production Workers*
	1983	*Single Unit*
	1984	*Total # Employees, Single Unit, Average # Production Workers*
	1985	*Total # Employees, Single Unit, Average # Production Workers*
	1986	*0*
1987	1987	*Single Unit*
	1988	*0*
	1989	*0*
	1990	*0*
	1991	*0*
1992	1992	*0*
	1993	*0*
	1994	*0*
	1995	*Single Unit*
	1996	*0*
1997	1997	*0*

[a] If a variable name is listed, there was a statistically significant difference between the means of inspected and non-inspected plants at the 10% level.
[b] A "0" indicates that all variables were balanced between the two samples in this year.

Table 6. DID estimates of plant-level changes in risks in response to receiving an OSHA inspection (dependent variable is annual plant-level fatality rates per 10,000 workers).[a]

	Estimated Coefficients (standard errors)[b]					
Variable Name	***Seven Industries***			***Full Sample***		
Turnover	0.026	0.008	0.049	0.035	0.024	0.043
	(0.171)	(0.168)	(0.202)	(0.089)	(0.089)	(0.104)
Productivity	0.197	0.215	0.208	0.074	0.079	0.076
	(0.160)	(0.160)	(0.171)	(0.059)	(0.060)	(0.064)
Number of Employees	0.069	-0.032	0.349	0.014	0.021	0.048
	(0.497)	(0.491)	(0.637)	(0.062)	(0.063)	(0.084)
Cost of Materials	-0.018	-0.009	0.003	-0.006	-0.005	-0.00004
	(0.025)	(0.027)	(0.031)	(0.004)	(0.005)	(0.005)
No. Production Workers	0.162	0.245	-0.144	-0.002	0.061	-0.049
	(0.612)	(0.588)	(0.748)	(0.153)	(0.158)	(0.188)
Single Unit Plant	0.217	0.235	0.185	0.117	0.137	0.130
	(0.282)	(0.281)	(0.291)	(0.133)	(0.133)	(0.140)
Programmed	-3.393***	-3.368***	-3.522***	-2.483***	-2.480***	-2.605***
Inspections (PI)	(0.691)	(0.712)	(0.751)	(0.371)	(0.380)	(0.405)
Fixed Effects Included:	Plant; Year	Plant; Year; Industry; Industry-by-Year	Plant; Year; Industry; Industry-by-State-by-Year	Plant; Year	Plant; Year; Industry; Industry-by-Year	Plant; Year; Industry; Industry-by-State-by-Year
Number of obs.	112,865	112,865	112,865	254,514	254,514	254,514
R-squared	0.001	0.002	0.028	0.001	0.002	0.028
Number of plants	28,071	28,071	28,071	65,326	65,326	65,326
F-statistic[c]	24.11	22.38	22.00	44.77	42.69	41.34

[a]Statistical significance at the 1%, 5%, and 10% level are represented by ***, **, and *, respectively.

[b] All models are estimated with standard errors clustered at the plant level that are robust to within-unit autocorrelation.

[c]The reported F-statistics test the restriction that *Programmed Inspections* has no effect on plant-level accident/fatality rates.

Table 7. DID estimates comparing wages at inspected and uninspected plants (dependent variable is hourly production worker wages).[a]

	Estimated Coefficients (standard errors)[b]					
Variable Name	***Seven Industries***			***Full Sample***		
Turnover	0.145**	0.143**	0.137**	0.138***	0.152***	0.153***
	(0.059)	(0.059)	(0.060)	(0.040)	(0.040)	(0.040)
Productivity	0.855**	0.847**	0.888***	0.608***	0.598***	0.614***
	(0.360)	(0.357)	(0.332)	(0.154)	(0.153)	(0.144)
Number of Employees	-0.442	-0.448	-0.484	-0.066	-0.034	-0.0007
	(0.725)	(0.716)	(0.727)	(0.229)	(0.223)	(0.214)
Cost of Materials	0.041**	0.033**	0.031**	0.012*	0.010	0.010
	(0.017)	(0.016)	(0.016)	(0.007)	(0.007)	(0.007)
No. Production Workers	-2.516***	-2.422***	-2.277***	-1.577***	-1.615***	-1.668***
	(0.732)	(0.736)	(0.747)	(0.374)	(0.370)	(0.354)
Single Unit Plant	0.255***	0.249**	0.245**	0.262***	0.266***	0.263***
	(0.099)	(0.099)	(0.099)	(0.064)	(0.064)	(0.065)
Programmed Inspections (PI)	-0.299***	-0.203***	-0.206***	-0.259***	-0.229***	-0.203***
	(0.074)	(0.074)	(0.075)	(0.052)	(0.052)	(0.054)
Fixed Effects Included:	Plant; Year	Plant; Year; Industry; Industry-by-Year	Plant; Year; Industry; Industry-by-State-by-Year	Plant; Year	Plant; Year; Industry; Industry-by-Year	Plant; Year; Industry; Industry-by-State-by-Year
Number of obs.	112,865	112,865	112,865	254,514	254,514	254,514
R-squared	0.213	0.218	0.237	0.192	0.200	0.222
Number of Plants	28,071	28,071	28,071	65,326	65,326	65,326

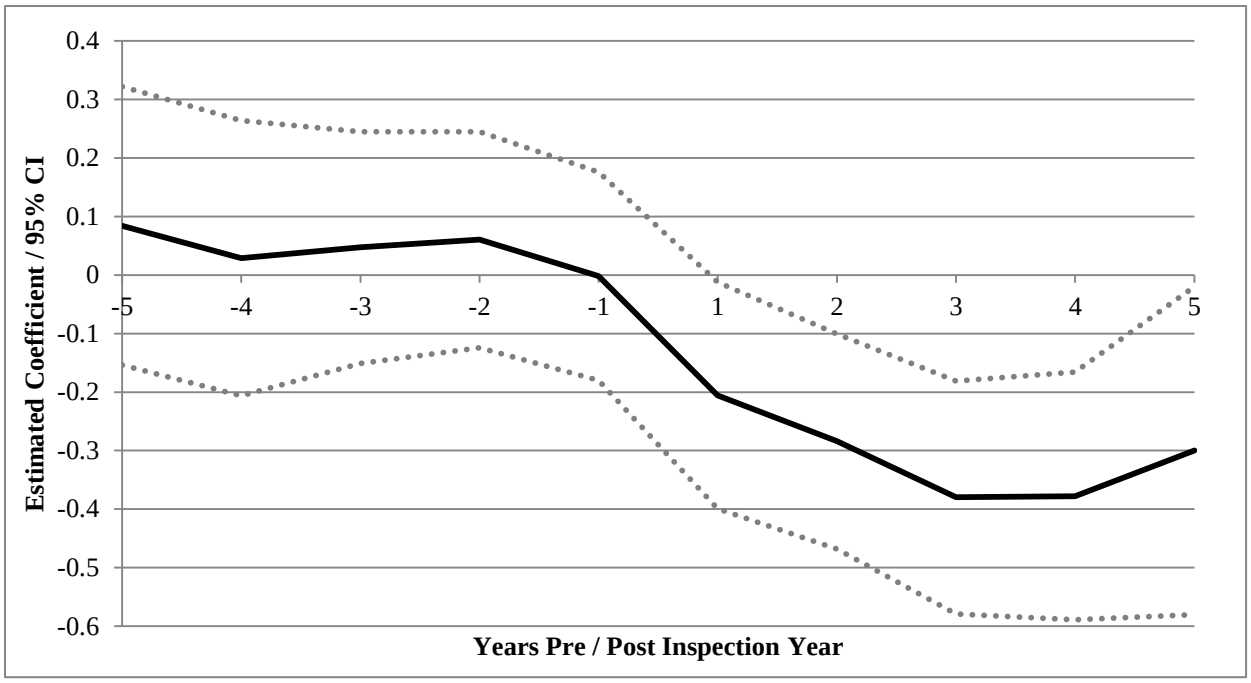
[a]Statistical significance at the 10%, 5%, and 1% level are represented by *, **, and ***, respectively.
[b] All models are estimated with standard errors clustered at the plant level that are robust to within-unit autocorrelation.

Table 8. Estimates of the VSL.[a]

Variable Name						
			Panel A: Difference in Differences Estimators			
	Seven Industries			***Full Sample***		
ρ point estimate (Table 7)	-0.299***	-0.203***	-0.206***	-0.259***	-0.229***	-0.203***
φ point estimate (Table 6)	-3.393***	-3.368***	-3.522***	-2.483***	-2.480***	-2.605***
Fixed Effects Included:	Plant; Year	Plant; Year; Industry; Industry-by-Year	Plant; Year; Industry; Industry-by-State-by-Year	Plant; Year	Plant; Year; Industry; Industry-by-Year	Plant; Year; Industry; Industry-by-State-by-Year
VSL (2011$)	3.49	2.39	2.31	4.12	3.66	3.09
[95% Confidence Interval]	[1.28 – 5.69]	[0.40 – 4.37]	[0.40 – 4.22]	[2.09 – 6.16]	[1.68 – 5.63]	[1.25 – 4.93]
			Panel B: Cross-Sectional, Uninstrumented Estimators (Dependent Variable is Wage)			
	2 Digit SIC Fatality Rates			***4 Digit SIC Fatality Rates***		
∂ point estimate (Equation 14)	2.027***	-0.142	-0.316***	0.176***	0.067***	-0.014
Fixed Effects Included:	Year	2 Digit SIC Industry; Year	Plant; Year	Year	2 Digit SIC Industry; Year	Plant; Year
VSL (2011$)	80.26	-5.62	-10.13	6.95	2.65	-0.45
[95% Confidence Interval]	[71.41 – 89.11]	[-13.23 – 1.99]	[-15.10 – -5.17]	[5.40 – 8.51]	[1.41 – 3.89]	[-1.08 – 0.18]
			Panel C: Cross-Sectional, Uninstrumented Estimators (Dependent Variable is ln(Wage))			
	2 Digit SIC Fatality Rates			***4 Digit SIC Fatality Rates***		
∂ point estimate (Equation 14)	0.204***	-0.002	-0.011*	0.019***	0.008***	-0.0002
Fixed Effects Included:	Year	2 Digit SIC Industry; Year	Plant; Year	Year	2 Digit SIC Industry; Year	Plant; Year
VSL (2011$)	78.06	-0.64	-4.55	7.30	3.02	-0.08
[95% Confidence Interval]	[71.49 – 84.63]	[-6.46 – 5.18]	[-9.24 – 0.15]	[5.98 – 8.61]	[2.45 – 3.58]	[-0.64–0.49]

[a]Statistical significance at the 1%, 5%, and 10% level are represented by ***, **, and *, respectively.

Figure 1: Event Study Analysis of Compensating Wages.[a]



[a]Results based on estimation of equation (3). The sample includes all inspected plants five years before and after treatment and uninspected plants matched within the same industry, state, and year.

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APPENDIX

Table A1: Characteristics by SIC Classification (1997).[1]

Variable Name	SIC 20: Food and Kindred Products	SIC 26: Paper and Allied Products	SIC 27: Printing and Publishing	SIC 28: Chemicals and Allied Products	SIC 32: Stone, Clay, and Glass Products	SIC 33 Primary Metal Industries	SIC 34 Fabricated Metal Products
Total Value of Shipments ($millions)	16.0	25.6	2.3	30.8	5.3	33.2	3.9
Production Workers' Payroll ($millions)	1.0	2.6	0.4	1.4	0.7	3.4	0.6
Number of Employees	55.8	97.9	19.5	65.5	30.7	119.6	28.3
Number of Production Workers	42.3	75.0	14.2	38.0	23.8	94.8	21.3
(% of total employees)	(76%)	(77%)	(73%)	(58%)	(78%)	(79%)	(75%)
Total Hours Worked By Production Workers (thousands)	84.9	159.5	27.2	79.3	48.9	202.5	42.9
Fatality Rate (per 100,000 workers)[2]	7.5	5.1	4.3	7.4	12.3	12.3	4.5
Injury Rate (per 100 workers)	19.8	12.2	10.4	10.7	17.2	20.6	19.1
Cost of Materials (millions)	9.8	13.7	0.9	14.3	2.3	19.6	1.8
Single Unit	75%	53%	92%	59%	64%	63%	88%
Productivity (in dollars)	188.76	160.61	83.75	388.80	108.39	164.07	90.77
Wages (in dollars)	11.27	16.23	14.03	17.95	14.19	16.94	13.55

[1] All data are reported in millions, unless otherwise noted. All data are from publicly available COM files for the selected manufacturing industries for the year 1997. The data is publicly downloadable via http://factfinder.census.gov/home/saff/main.html?_lang=en (last accessed Nov. 2011).

Table A2: DID Estimates Comparing Inspected and Uninspected Plant Wages Prior to the Inspection Year.[a]

	Estimated Coefficients (standard errors)	
Variable Name[b]	***Seven Industries***	***Full Sample***
Turnover	0.126***	0.181***
	(0.047)	(0.030)
Productivity	2.556***	1.49***
	(0.557)	(0.412)
Total # Employees	-0.084	0.481***
	(0.584)	(0.184)
Cost of Materials	0.027	0.004
	(0.022)	(0.007)
Avg. # Production Workers	-1.494**	-1.802***
	(0.751)	(0.297)
Single Unit	0.203**	0.108*
	(0.082)	(0.056)
Programmed Inspections[c]	0.101	0.065
	(0.085)	(0.061)
Number of obs.	75,575	166,794
R-squared	0.390	0.341
Number of Plants	19,365	42,658

[a]Statistical significance at the 10%, 5%, and 1% level are represented by *, **, and ***, respectively. Although not reported, each model presented also includes a full set of plant and year fixed effects as indicated in equation (6).

[b] Wages and plant-specific characteristics are from the period 1977 to 1987.

[c] Firms are categorized as inspected or not based on inspection status at 10 years post-wage and plant characteristics data (i.e., during the period 1987 to 1997).

Table A3: DID Estimates Comparing Inspected and Uninspected Plant Natural Log of Wages.[a]

	Estimated Coefficients (standard errors)[b]					
Variable Name	**Seven Industries**			**Full Sample**		
Turnover	0.004	0.004	0.003	0.006**	0.007**	0.007**
	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)	(0.003)
Productivity	0.064**	0.064**	0.066***	0.045***	0.045***	0.046***
	(0.026)	(0.026)	(0.024)	(0.011)	(0.011)	(0.010)
Total # Employees	-0.014	-0.015	-0.022	0.008	0.009	0.011
	(0.037)	(0.037)	(0.038)	(0.012)	(0.012)	(0.012)
Cost of Materials	0.001	0.001	0.001	0.0006	0.0005	0.0005
	(0.001)	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)
Avg. # Production Workers	-0.179***	-0.177***	-0.168***	-0.114***	-0.118***	-0.125***
	(0.042)	(0.042)	(0.042)	(0.023)	(0.023)	(0.023)
Single Unit	0.026***	0.025***	0.025***	0.027***	0.028***	0.027***
	(0.008)	(0.008)	(0.008)	(0.005)	(0.005)	(0.005)
Programmed Inspections	-0.020***	-0.016***	-0.016***	-0.018***	-0.017***	-0.015***
	(0.006)	(0.006)	(0.006)	(0.004)	(0.004)	(0.004)
Fixed Effects Included in Models:	Plant; Year	Plant; Year; Industry; Industry-by-Year	Plant; Year; Industry; Industry-by-State-by-Year	Plant; Year	Plant; Year; Industry; Industry-by-Year	Plant; Year; Industry; Industry-by-State-by-Year
Number of obs.	112,865	112,865	112,865	254,514	254,514	254,514
R-squared	0.238	0.241	0.260	0.221	0.225	0.222
Number of Plants	28,071	28,071	28,071	65,326	65,326	65,326
VSL (2011$)	2.94	2.30	2.20	3.34	3.22	2.81
[95% Conf. Interval]	[0.88 – 5.00]	[0.34 – 4.25]	[0.31 - 4.09]	[1.49 – 5.18]	[1.37 – 5.07]	[1.06 - 4.56]

[a]Statistical significance at the 10%, 5%, and 1% level are represented by *, **, and ***, respectively. Although not reported, each model presented also includes a full set of plant and year fixed effects as indicated in equation (4).
[b] All models are estimated with standard errors clustered at the plant level that are robust to within-unit autocorrelation.